

PHYSICS OF COMPLEX SYSTEMS

LECTURE AND TUTORIALS – PROF. DR. HAYE HINRICHSSEN – MAXIMILIAN SCHÄFFERER – SS 2026



Joseph Liouville 1809-1882

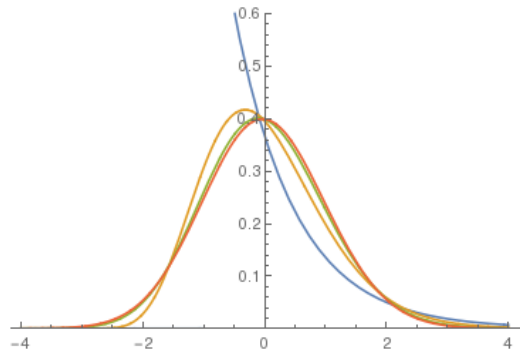
EXERCISE 3.1: CENTRAL LIMIT THEOREM (CLT)

(6P)

Let $X_1, X_2, \dots, X_N$ be statistically independent and identically distributed random variables with the probability density function

$$p_X(x) = \begin{cases} e^{-x} & \text{if } x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

With this exercise we would like to demonstrate the CLT for $Z_N = \sum_{i=1}^N X_i$ in the limit $N \rightarrow \infty$.



- (a) Show that the N -fold convolution product of $p(x)$ is given by (1P)

$$p_{Z_N}(z) = p_X^{*N}(z) = \underbrace{(p_X * p_X * \dots * p_X)}_{N \text{ times}}(z) = \frac{z^{N-1}}{(N-1)!} e^{-z} \quad (z \geq 0).$$

- (b) Check the norm and compute the mean and the variance of $p_{Z_N}(z)$. (1P)
- (c) Standardize the distribution by shifting and rescaling Z_N in such a way that the new random variable $\tilde{Z}_N$ has a probability density $p_{\tilde{Z}_N}(z)$ with unit norm, zero mean, and unit variance (see figure). (1P)
- (d) Show that the cumulant-generating function $K_{\tilde{Z}}(t)$ of the standardized probability density $p_{\tilde{Z}_N}(z)$ is given by (1P)

$$K_{\tilde{Z}}(t) = \frac{1}{2}N \ln N - t\sqrt{N} - N \ln(\sqrt{N} - t).$$

- (e) Compute all cumulants κ_n and show that only κ_2 survives in the limit $N \rightarrow \infty$. What does it mean? (2P)

$\Rightarrow$ PLEASE TURN OVER

EXERCISE 3.2: VECTOR NOTATION AND LIOUVILLE OPERATOR (6P)

Consider a continuous-time Markov process (see lecture notes) with the master equation

$$\dot{P}_c(t) = \sum_{c'} P_{c'}(t) w_{c' \rightarrow c} - P_c(t) \sum_{c'} w_{c \rightarrow c'} \quad \Leftrightarrow \quad |\dot{P}\rangle = -\mathcal{L}|P\rangle.$$

- (a) Prove that the matrix elements of the Liouville operator $\mathcal{L}$ are given by (3P)

$$\langle c' | \mathcal{L} | c \rangle = -w_{c \rightarrow c'} + \delta_{c,c'} \sum_{c''} w_{c \rightarrow c''}$$

- (b) Consider a system with three configurations 1, 2, 3 and the transition rates

$$w_{1 \rightarrow 2} = w_{3 \rightarrow 2} = 3, \quad w_{2 \rightarrow 1} = 1, \quad w_{1 \rightarrow 3} = \frac{1}{2}, \quad w_{2 \rightarrow 3} = 2,$$

in units of 1/time (all other rates are zero). Determine the matrix of the Liouville operator. (1P)

- (c) Calculate the so-called *zero vectors*, i.e., the left and right eigenvectors to the eigenvalue zero. (1P)
- (d) Normalize the right zero vector in order to interpret it as a probability distribution which describes the stationary state of the system. (1P)

($\Sigma = 12P$)

Please submit your solution electronically as a single PDF file until Wednesday, May 06, 2026, via Wue-Campus.